
Mapping compatibility boundaries for magnetized cross-beam energy-transfer theory

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Abstract

Stationary models of magnetized cross-beam energy transfer need a parameter-resolved test against finite, evolving plasmas. We reconstruct the Maxwellian susceptibility tensor and map gain compatibility across magnetic-field strength, field orientation, and electron temperature, fixing the tolerance before evaluating residuals. At a common detuning, the 70 archived configurations comprise 41 discrepant, 19 unresolved, and 10 compatible cases; only 3 compatible cases have nonzero, nonduplicated sensitivity information. A sensitivity-supported field transition is bracketed between 12.5 and 20 MG in 1.2 keV hydrogen with orthogonal polarization. 12 new EPOCH calculations extend the comparison to carbon at $B = 10$ MG and $\phi_B = 30^\circ$, with three-seed tolerance transitions bracketed over 0.4 to 0.455 keV after a disclosed adaptive refinement. The compatible 455 eV point nevertheless shows late gain evolution (Holm-adjusted $p = 0.0323$, conditional on a three-seed normal sampling model). The carbon reference already disagrees at zero field, where its reconstructed ion-acoustic relaxation time of 56.11 ps exceeds the 3.6 ps observation time. Particle, field, and independently measured-state diagnostics leave the physical mechanisms unattributed. The map locates sampled tolerance crossings and identifies the equilibrium, temporal, and numerical controls required before such crossings can guide a choice between stationary theory and kinetic simulation.

1 Disagreement and the conditions for a kinetic test

Cross-beam energy transfer (CBET) redistributes laser power through a driven plasma density perturbation. This redistribution can improve indirect-drive symmetry (7) or reduce direct-drive coupling (3). Magnetization changes both electromagnetic polarizations and the low-frequency response. Its relevance extends beyond an applied field: in integrated implosion calculations, transport, equilibrium profiles, and the empirical treatment of CBET saturation change together (17). A useful theory-validation boundary must distinguish these effects.

The kinetic theory of Moody and Shi (8) retains magnetized electron and ion susceptibilities, together with an approximate ponderomotive drive. Its physical content includes additional magnetized plasma resonances. Their appearance alone is consequently not evidence against that theory. Likewise, comparisons with PIC must establish compatible interaction geometry, sufficiently stationary plasma conditions, controlled losses, and adequate numerical resolution before assigning a discrepancy to the drive approximation or nonlinear ion motion. The need to solve an initial-boundary-value problem rather than compare unrelated growth measures was demonstrated for magnetized laser backscattering by Shi (11). Nonlinear CBET studies independently show that trapping, ion-wave evolution, pump depletion, and secondary scattering can change the gain in time (9; 18).

Here we ask how much of a quantitative validity boundary can be identified in finite, initially Maxwellian plasmas. Our contribution is an auditable parameter map combining a reconstructed kinetic response, unaltered archived measurements, and independently executed EPOCH calculations. We separate a fixed numerical tolerance from statistical confidence and from physical attribution. This

distinction matters especially when magnetic suppression makes both the predicted and measured transfer small: numerical compatibility can then occur without a stringent test of the coupling physics.

The published 2026 kinetic and PIC studies (8; 14) make this distinction consequential: polarization mixing and magnetized resonances are now represented explicitly, while quantitative comparisons still depend on the preparation and evolution of the plasma. Established susceptibility and magnetized coupling methods (15; 10; 6) underlie our reconstruction. The contribution lies in determining which parameter-dependent conclusions survive those comparison limits.

2 Kinetic response and the measured energy

We use SI units, a homogeneous collisionless equilibrium, and isotropic Maxwellians with the input densities and temperatures. Let $\omega_p > \omega_s$ denote the pump and seed laser frequencies, $\omega = \omega_p - \omega_s$, and $\mathbf{q} = \mathbf{k}_p - \mathbf{k}_s$. For each species a , the signed gyrofrequency is $\Omega_a = q_a B / m_a$ and $v_a = (2eT_a/m_a)^{1/2}$, with T_a in electronvolts. The longitudinal projection of the magnetized susceptibility is

$$\chi_{aL} = \frac{2\omega_{pa}^2}{q_{\parallel}^2 v_a^2} \left[1 + \frac{\omega}{q_{\parallel} v_a} \sum_{n=-N_a}^{N_a} \Gamma_n(b_a) Z\left(\frac{\omega - n\Omega_a}{q_{\parallel} v_a}\right) \right], \quad b_a = \frac{q_{\perp}^2 v_a^2}{2\Omega_a^2}, \quad (1)$$

where $\Gamma_n(b) = e^{-b} I_n(b)$ and Z is the Landau-contour plasma dispersion function (2; 15). We evaluate Z through SciPy's complex error function and use scaled Bessel functions. The cutoff is $N_a = \max[10, \lceil 12\sqrt{b_a} + 1 \rceil]$. The unmagnetized limit is evaluated directly, avoiding a diverging harmonic sum.

The gain calculation retains the full tensor, not only Eq. (1). In the notation of Moody and Shi (8),

$$\mathbf{D} = \mathbf{I} + \chi_e + \chi_i + \mathbf{N}\mathbf{N}^T - N^2\mathbf{I}, \quad \mathcal{R} = \mathbf{q}^T [\chi_e - \chi_e \mathbf{D}^{-1} \chi_e] \mathbf{q}, \quad \mathbf{N} = \frac{c\mathbf{q}}{\omega}. \quad (2)$$

Linear solves implement the inverse action. The high-frequency electromagnetic modes follow from the cold dielectric tensor after elimination of the longitudinal field. Their transverse eigenvectors are used in the coupling and PIC mode analysis. The model fixes the nominal propagation directions; it does not solve refraction or the numerical optical dispersion imposed by the open-boundary injection. The drive retains the potential term evaluated at the mean laser frequency and omits the magnetization and time-derivative terms, as in the reconstructed model (5; 8). The largest longitudinal laser-field fraction in the evaluated configurations is 0.0054, which checks the transverse approximation but does not bound the error in a resonant gain.

The four pump–seed mode pairs are coupled using the finite spherical phase average

$$S(x) = 3 \frac{\sin x/x - \cos x}{x^2}, \quad x = |\Delta\mathbf{k}|L/2, \quad S(0) = 1. \quad (3)$$

We solve the resulting two linear equations for the seed index increments δn_j directly. This follows the published finite-interaction prescription; a sphere of diameter L is not the elongated open PIC box. No gain normalization, loss offset, field-dependent correction, or effective temperature is fitted to PIC. In particular, the reported baseline is an independently computed stationary reconstruction, not a numerical extraction of published theory curves. Appendix A specifies the tensor and coupling implementation and its scope.

The observable is an integrated log-energy gain. An exponential intensity exponent $G_j = -2(\omega_s/c) \text{Im}(\delta n_j)L$ corresponds to

$$F(G) = \frac{e^G - 1}{G}, \quad F(0) = 1, \quad g_V = \log \left[\sum_j w_j F(G_j) \right], \quad \sum_j w_j = 1, \quad (4)$$

where w_j is the injected seed's modal energy fraction. This conversion is essential: an intensity exponent and an integrated energy gain are different quantities. We also port the authors' two-adjacent-wavevector formula (14; 12) as a finite-box sensitivity calculation. It remains separate from the primary continuous-frequency prediction because a temporally driven open boundary does not enforce a periodic-box spectrum.

We locate longitudinal dielectric roots of $1 + \chi_{eL} + \chi_{iL} = 0$ from four fixed complex initial guesses around the fluid sound frequency, retaining a damped positive-frequency root with residual below 10^{-6} and closest real frequency to that estimate. This is a diagnostic selector, not a continuation proof or an enumeration of every magnetized branch. The real-frequency gain always uses Eq. (2). Projection of the implemented tensor onto $\mathbf{q}$ reproduces Eq. (1) to 2.35×10^{-14} relative error; increasing the harmonic cutoff multiplier from 12 to 16 changes the demanding convergence case by 2.33×10^{-17} . The unmagnetized tensor and electrostatic calculations differ by at most 3.15×10^{-14} in the computed log-energy gain.

3 Plasma configurations and comparison protocol

The archived data (13) contain 27 distinct carbon settings and 43 hydrogen settings. Carbon uses $n_e = 3.5 \times 10^{20} \text{ cm}^{-3}$, $T_e = 4T_i = 0.8 \text{ keV}$, fully stripped ions, a 30° crossing angle, and equal pump and seed intensities $5 \times 10^{14} \text{ W cm}^{-2}$. Its field lies in the crossing plane, with $\phi_B = -13^\circ, 30^\circ, 75^\circ$, and the core field values are $B = 0, 2, 5, 10, 50 \text{ MG}$. The additional archived 80 MG cross-polarized point is retained. Zero field has no physical field azimuth and is counted once per polarization.

Hydrogen uses $n_e = 3 \times 10^{20} \text{ cm}^{-3}$, $T_e = 0.4 \text{ or } 1.2 \text{ keV}$, $T_i = T_e/4$, a 24° crossing angle, and $(\theta_B, \phi_B) = (45^\circ, 15^\circ)$. Ionic masses follow the input decks: $m_C = 12(1837.2m_e)$ and $m_H = 1836m_e$. Inspection of the hydrogen decks gives $10^{15} \text{ W cm}^{-2}$ for both beams, 10 cells per wavelength, and 5 particles per cell; these differ from the carbon reference's 20 cells per wavelength and 20 particles per cell. The available field values and every configuration are enumerated in Table 4. This is a nonfactorial design: temperature, species, intensity, geometry, and estimator change together between archive families. Temperature therefore cannot be interpreted as an isolated causal variable from the archive alone.

A time-stamped design fixed the field, azimuth, and temperature ranges before any gain rows were read or residuals computed. It additionally specified carbon temperature settings at 0.4 and 1.2 keV with $B = 10 \text{ MG}$ and $\phi_B = 30^\circ$, conditional on successful numerical validation. The reference and new configurations use the same declared reduction: a $250\lambda_0 \times 10\lambda_0$ box, $\lambda_0 = 0.351 \mu\text{m}$, 12 cells per wavelength, 12 total particles per cell, and the supplied laser/CPML configuration. A finer 20-cell, 20-particle validation alternative was fixed in the same design. A separately dated adaptive refinement adds 0.455 keV after the first completed 0.4 keV realization; it uses independent new seeds and is distinguished from the original design in Sec. 6. The reference gate and its executed outcome are reported there; no failed reference is relabeled as a physical boundary.

Measurements and uncertainty. Carbon's primary archive values are the real-space eigenmode-analysis files, with the original Fourier-power estimates retained as an estimator audit. Hydrogen has only the Fourier-power estimates. In the new runs we port the supplied Gaussian Fourier filter, transverse integration fractions $f_y \in \{1, 0.75, 0.5\}$, second longitudinal filter, phase shift, and eigenmode energy decomposition (12). We compare the late snapshots near 3.0 and 3.6 ps with the seed-transit snapshot just before the delayed pump. The mean of the six log-energy ratios is the central value; their full maximum-minus-minimum range is retained. No negative background is subtracted. The archive does not contain the individual six values, so its time and window contributions cannot be separated retrospectively.

The primary comparison uses $\Delta\lambda = 0.9 \text{ \AA}$. A missing exact archive row is linearly interpolated between bracketing detunings, including interpolation of its reported range; no extrapolation is used. All available detuning rows are also evaluated and retained. With the preselected floor $g_* = 0.5$, define

$$D(g_V, g_P) = \frac{|g_V - g_P|}{\max(|g_P|, g_*)}, \quad \tau = 0.30. \quad (5)$$

Thus the tolerance becomes an absolute 0.15 in the low-gain regime. For a reported spread u , we compute the exact extrema of D over $g_P \in [\bar{g}_P - u, \bar{g}_P + u]$. A configuration is *compatible* if the maximum is at most τ , *discrepant* if the minimum exceeds τ , and otherwise *unresolved*. Using a full spread on each side is deliberately conservative relative to the observed six-value range. It is not a confidence interval for unmeasured systematic errors.

Statistical interpretation. The time/window variants are correlated sensitivity checks, not independent realizations. We do not convert them into Gaussian standard errors or cellwise p values. Repeated late-time indices and zero spreads are explicitly marked illustrative. The prespecified physical hypotheses concern drive validity at larger $|\Omega_e|/\omega_p$, nonlinear ion response, and enhanced discrepancy near perpendicular propagation; a separate hypothesis concerns comparison validity. Intensity is not independently varied here, so nonlinear-ion-drive attribution is not identifiable. The registered inferential family, conditional fresh-cell tests, and Holm correction at family-wise level 0.05 (4) are reported with the executed results. No false-discovery-rate procedure is substituted for family-wise control.

4 The finite-plasma compatibility map

Figure 1 combines the archived configurations with the new temperature line discussed in Sec. 6. Of 70 archived configurations, 41 are discrepant, 19 unresolved, and 10 compatible. The sensitivity-supported subset contains 22, 19, and 3 in those respective categories. The remaining 26 configurations have repeated late snapshots or zero spread and are illustrative; they are not promoted to established agreement or disagreement. Table 4 gives every central prediction, measurement, residual, and attribution status. The full detuning comparisons and the alternative archive estimator are retained in the machine-readable results. The sampled locations of the tolerance transitions are more informative than the counts alone. For 1.2 keV hydrogen with z - y polarization and the fixed hydrogen geometry, the last compatible field is 12.5 MG and the next definitive discrepant field is 20 MG. The intervening 17.5 MG point is unresolved. This brackets a change in the finite-sensitivity decision, not a confidence bound on a continuous critical field. Its physical mechanism remains unattributed. The low-temperature hydrogen transitions are illustrative: z - y changes from compatible at 5 MG to discrepant at 7.5 MG; z - z changes from discrepant at 20 MG to compatible at 22.5 MG. Their zero reported ranges do not establish precision. No carbon archive point is fully compatible over its documented sensitivity interval. The few supported compatible hydrogen points are all near zero gain, where the declared absolute floor makes the test least stringent.

The key obstruction precedes magnetization. At the carbon zero-field, parallel-polarization reference, the raw PIC gain is 1.121 ± 0.266 , whereas the reconstruction gives 0.0235. Its nominal discrepancy is $D = 0.979$. Thus there is no quantitatively validated unmagnetized starting regime from which these data establish the first failure encountered as the field rises. A field threshold inferred by labeling these residuals alone would confound a pre-existing mismatch with magnetized physics.

Figure 2 separates this absolute mismatch from the qualitative magnetic trend. The signs of the endpoint changes agree in 5 of the 6 angle/polarization comparisons, including suppression of parallel-polarization gain and activation of cross-polarized transfer in the nearly parallel plasma-wave geometry. However, the reconstruction does not reproduce monotonic suppression at every field angle. In particular, the nearly perpendicular z - z prediction increases between the zero-field and high-field endpoints at the chosen detuning, contrary to the PIC central values. A claim of comprehensive qualitative reproduction would therefore be unwarranted. Agreement of a trend's sign is insufficient to validate its amplitude or spectral location.

The unmagnetized spectrum clarifies why a fixed-frequency comparison is delicate (Fig. 3). The reconstructed carbon ion-acoustic root corresponds to $\Delta\lambda = 1.235 \text{ \AA}$ and a relaxation time $1/|\text{Im } \omega| = 56.11 \text{ ps}$. The measured spectrum is broad, whereas the initial-Maxwellian stationary response is sharply resonant. The large resonant peak is the ideal undepleted-pump prediction, not a sustainable saturated gain. It is retained to expose the spectral mismatch; nonlinear saturation would limit its amplitude. The discrete-mode sensitivity prescription does not resolve this identification problem: its central estimate changes 4 of the 70 classifications, with 0 changes among the sensitivity-supported configurations. The changed labels belong to illustrative low-temperature hydrogen cases. Its dependence on which pair of adjacent spatial modes is excited is itself strong. We retain those outcomes rather than selecting the more favorable prediction for each configuration.

5 Parameter dependence and kinetic interpretation

The prespecified mechanisms do not acquire support merely because a point is colored discrepant. The drive hypothesis predicts a deterioration associated with increasing electron magnetization at

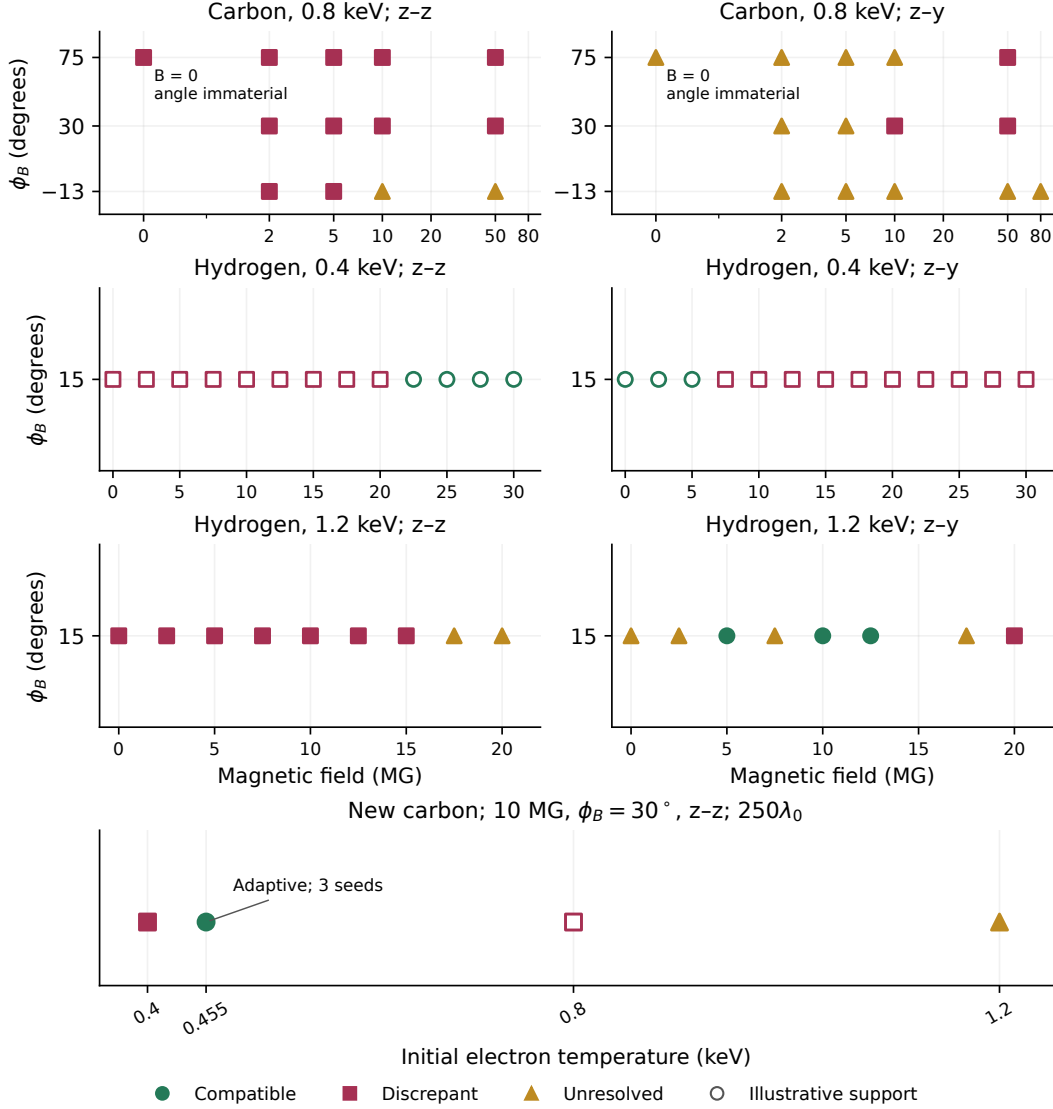


Figure 1: Compatibility of the stationary reconstruction at $\Delta\lambda = 0.9 \text{ \AA}$. Colors and shapes denote Eq. (5); hollow symbols mark illustrative support. The upper six panels contain every distinct archive configuration. Carbon field axes use a symmetric-log scale and hydrogen axes a linear scale. The hydrogen rows differ from carbon in species, intensity, resolution, and field geometry; they do not isolate temperature. Zero-field azimuth is immaterial. The bottom panel adds all new fixed-geometry carbon temperatures: filled points require three seeds and passing measurement checks; an exclamation mark would denote a failed check. The adaptive point is identified explicitly. Reference-calibration runs are tabulated separately. Compatibility is a sensitivity decision, not a confidence statement about omitted physics.

the laser frequency. The largest $|\Omega_e|/\omega_p$ in the archive is 0.262, and the largest beat-to-cyclotron ratio at nonzero field is 0.039. These ratios delimit the sampled drive window; they are not calibrated error estimates. The pronounced zero-field discrepancy already prevents identification of a first magnetized failure from that ordering. In matched carbon comparisons, the median change in D from nearly parallel to nearly perpendicular plasma-wave propagation is -0.356. This descriptive effect is opposite to the proposed increase in discrepancy near perpendicular propagation. It neither proves that perpendicular propagation is benign nor identifies a missing resonance: the tensor model already retains linear cyclotron harmonics. The hypothesis of nonlinear ion ponderomotive response

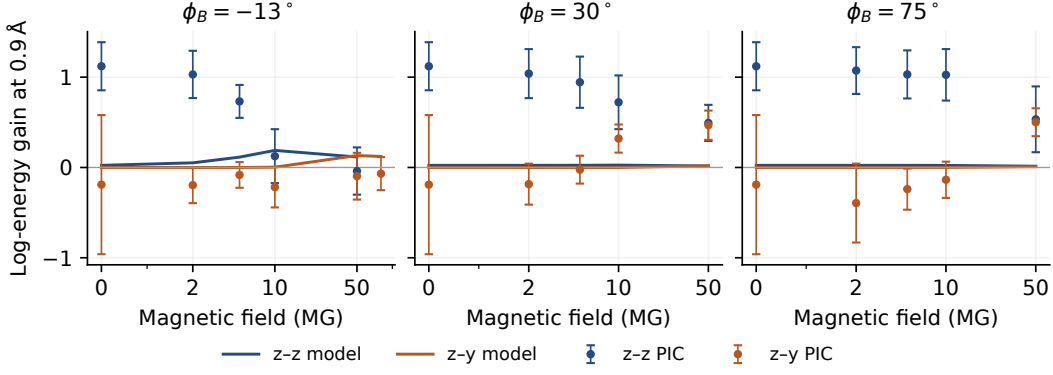


Figure 2: Carbon gain at the fixed detuning. Markers and full sensitivity ranges are the supplied measurements; curves join independent model evaluations. Both polarizations and every available field point are included. The zero-field measurements are reused visually across panels, but are counted only once per polarization in all summaries. The common vertical scale exposes the absolute discrepancy.

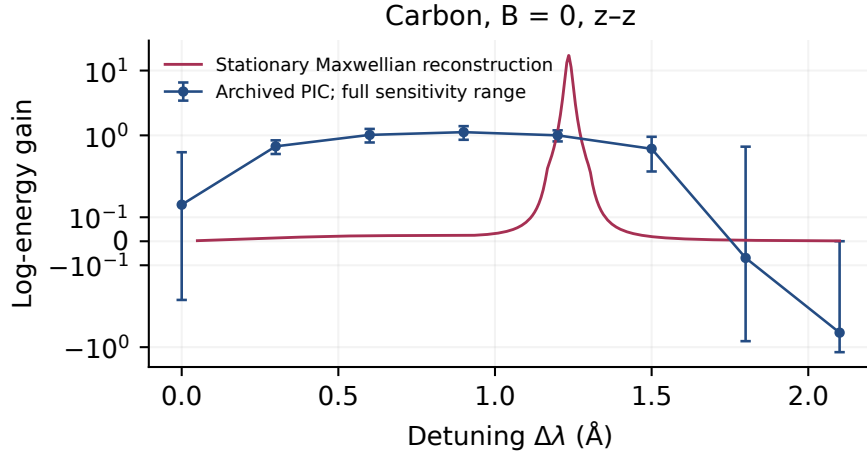


Figure 3: The carbon reference spectrum, including all supplied total-gain detunings. The theoretical curve is evaluated on a dense frequency mesh with the independently computed dielectric root included explicitly. The symmetric-log ordinate displays both the narrow theoretical peak and negative measured values without clipping either. This is a spectral discrepancy, not evidence that a magnetic-field approximation has failed.

cannot be isolated because laser intensity is fixed within each plasma family and confounded between families.

Comparison-validity mechanisms remain plausible but are not interchangeable with physical failure of the theory. The zero-field orthogonal-polarization central values include negative gains, for which a loss-free CBET model gives zero; their sensitivity ranges prevent a precise loss calibration. More generally, the processed archive cannot recover local evolving distributions, pump depletion, or separate time/window contributions. All archive mechanisms are therefore labeled *unattributed*. Appendix D audits the frequency orderings, selected-root damping, and available controls separately at every reported transition. The three original inferential entries are unstable from these correlated summaries and are conservatively assigned $p = 1$ for family accounting, not reported as measured null-hypothesis probabilities.

The long carbon relaxation time motivated an additional temporal hypothesis after inspecting the unmagnetized solver and archive reference. Its post-inspection origin is recorded explicitly. Before any completed new PIC gain was available, we fixed a prospective test on the two new temperature

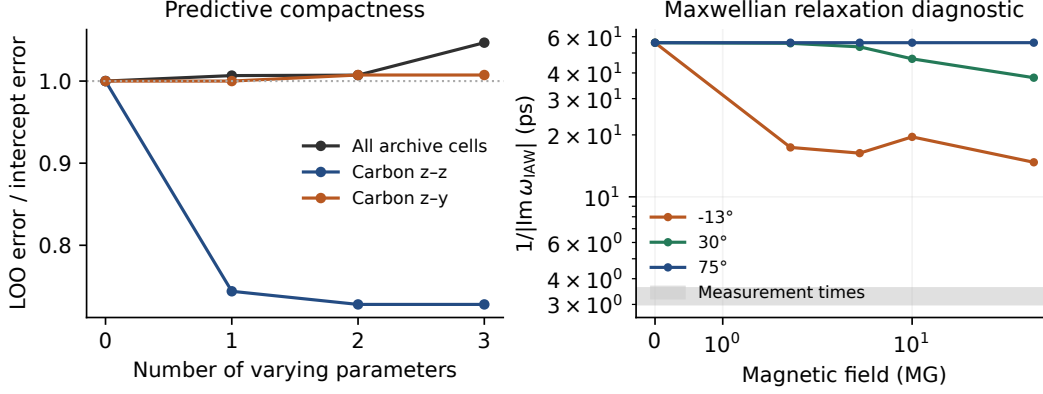


Figure 4: Left: best leave-one-configuration-out prediction error at each parameter count, normalized to the intercept-only error. Increasing complexity does not improve pooled prediction. Right: the selected longitudinal sound-frequency root’s relaxation time in carbon; the band marks the measurement times, not an uncertainty interval. Multiple roots can coexist near perpendicular propagation, so the diagnostic selector is not a unique branch identification.

configurations: average the three window gains within each seed, compute $g(3.6 \text{ ps}) - g(3.0 \text{ ps})$, and test its mean across independent seeds. These two tests join the three original entries in one Holm family. The adaptive temperature refinement described below adds a third fresh-cell temporal test, fixed before its independent realizations; the final family therefore has six entries. A small damping-rate–time product is only a warning about equilibration: convective escape and changing distributions can also set the response time. It is not a causal attribution by itself.

To quantify compactness, we fit all additive subsets of $\log(1+B)$, categorical azimuth, and categorical temperature to the continuous discrepancy D . A fixed ridge penalty of one is used after training-fold standardization; the intercept is unpenalized. Leave-one-configuration-out prediction follows standard cross-validation (16). At zero field, azimuth indicators vanish. Polarizations and plasma families are also examined separately; no held-out response is used to select a threshold or change the features. The errors compare this fixed model family on the same data; the selected minimum has no independent test-set evaluation.

The pooled intercept-only model has the smallest prediction error, 0.1068. For carbon z - z , the field-plus-azimuth model reduces error from 0.0865 to 0.0629, an improvement of 27.2%. No comparable improvement holds for the pooled sample or carbon z - y (Fig. 4). The archive therefore does not support a common compact boundary. Temperature’s failure to improve this predictor is not evidence of physical inertness: the nonfactorial design makes that interpretation unavailable. The modestly compact parallel-polarization subset is a description of residuals, not a transferable ICF validity surface.

6 New finite-plasma calculations

The reduced calculation is calibrated against the archived carbon z - z references at $B = 0$ and 50 MG, with $\phi_B = 75^\circ$ (14; 1). We invert $g = \log F(G)$ for the archived exponent, multiply G by the interaction-length ratio, and reconvert to g before comparison. This assumes exponential spatial growth solely for transferring the reference amplitude between lengths; it is not an independent validation of that assumption. Pump injection is delayed by a seed transit, so reducing the length also changes the elapsed time after pump startup at fixed observation times. The prespecified gate requires either a relative error no larger than 50% in the length-normalized exponent or an absolute difference no larger than 0.30 in g , spectral signal-to-background power ratio above one at each stored time from 3.0 through 3.6 ps, and a late sensitivity range no larger than 0.40. The high-field lower endpoint must not exceed the zero-field upper endpoint. This last check excludes a resolved reversal; it is weaker than demonstrating monotonic suppression. The coarse pair passed the complete gate. The same selected domain, resolution, particle loading, and measurement protocol were then used for every temperature run. The high-field case passes through the absolute-error allowance, although

Resolution	B (MG)	New $g \pm u$	Scaled reference	$ \Delta g $	SNR	Gate
12 cells, 12 ppc	0	0.661 ± 0.218	0.523	0.138	154.7	pass
12 cells, 12 ppc	50	0.546 ± 0.201	0.257	0.289	146.5	pass

Table 1: Executed reference validation. Every attempted numerical configuration is retained. “Pass” in a row refers to its amplitude, spectral, and stability tests; selection also requires the paired trend check. The tolerance was fixed before measuring these gains.

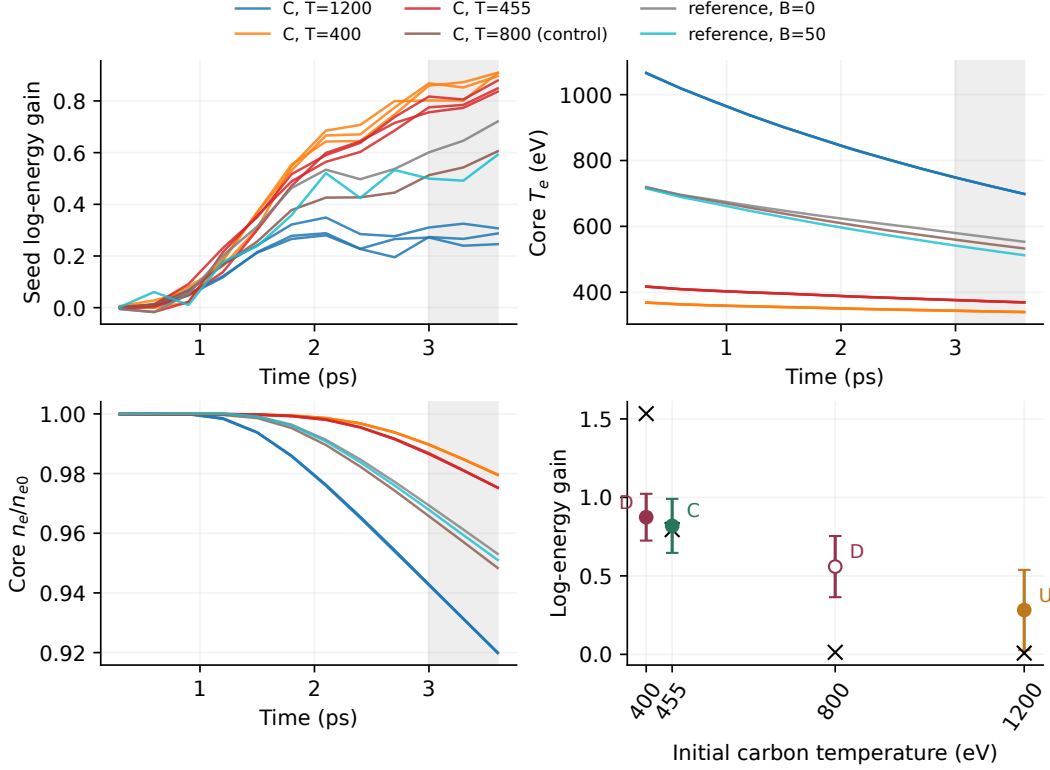


Figure 5: All completed EPOCH calculations. The gain traces average the three transverse windows within each snapshot; separate RNG seeds remain separate curves. Temperatures are density-weighted deposited moments, and density is a spatial mean, over the central 80% of each direction. The initialized temperature estimator is audited in Appendix C.1. These diagnostics are measured independently of gain. The gray region spans the two late measurement times. In the lower-right panel, circles and full sensitivity ranges show PIC and crosses show the initial-state stationary prediction. Hollow circles identify illustrative or invalid comparisons; C, D, and U give the fixed compatibility classification, with an exclamation mark for a failed measurement prerequisite.

its length-normalized exponent error exceeds the relative allowance. The overlapping sensitivity intervals support only the declared non-reversal criterion. This is a limited reference calibration and does not demonstrate numerical convergence at every new temperature. The new carbon temperature line holds $B = 10$ MG, $\phi_B = 30^\circ$, density, intensity, species, and interaction length fixed. The two originally planned endpoint temperatures use three independent base seeds each; the adaptive refinement uses a separate set of three new seeds. The 0.8 keV control was added after archive inspection but before reading a completed new gain to eliminate an otherwise confounded length comparison; it has one seed and remains illustrative. The refinement at 0.455 keV was selected after the first 0.4 keV realization: a theory-only scan at 440, 445, ..., 475 eV chose the prediction closest to that observed gain. This chooses a new experimental setting rather than fitting the baseline at an observed setting. Its prediction and three independent validation seeds were fixed before its PIC outcomes. It targets a nominal agreement region and therefore cannot be treated as a randomly sampled or originally registered location. All original configurations remain included. Table 2 reports

T_e (keV)	Seeds	g_V	$g_P \pm u$	Seed SD	Classification
0.4	3	1.534	0.874 ± 0.149	0.0165	discrepant
0.455	3	0.794	0.819 ± 0.172	0.0266	compatible
0.8	1	0.013	0.559 ± 0.195	one seed	discrepant
1.2	3	0.008	0.283 ± 0.256	0.0251	unresolved

Table 2: Fresh temperature configurations at the same reduced numerical setting. Every three-seed configuration includes all its realizations; the 0.8 keV control is illustrative, and 0.455 keV is the disclosed adaptive refinement. Classification uses the original tolerance and denominator floor without modification.

T_e (keV)	Mean late drift	Seed SD	Raw p	Holm p	Decision
0.4	0.0613	0.0402	0.1186	0.5929	not rejected
1.2	-0.0050	0.0199	0.7028	1	not rejected
0.455	0.0715	0.0091	0.00539	0.03234	reject

Table 3: Family-wise corrected temporal tests on fresh configurations. A rejection concerns a comparison-validity prerequisite, not the identity of omitted kinetic physics. Failure to reject would not establish stationarity, and a statistically nonzero drift need not exceed the chosen gain-error tolerance.

the range pooled over all seed/time/window values. The standard deviation across seed means is also shown; it is distinct from the sensitivity range and is not substituted for it. The fresh temperature line changes from discrepant at 0.4 keV to compatible at 0.455 keV. Across these endpoints the model gain changes by -0.739, while the PIC central gain changes by -0.055. These are descriptive differences, not additional significance tests. Both definitive endpoints have three-seed sensitivity support. The interval brackets sampled tolerance labels only; it does not locate a unique continuous temperature threshold or identify a physical failure mode. The fresh temperature line changes from compatible at 0.455 keV to discrepant at 0.8 keV. Across these endpoints the model gain changes by -0.781, while the PIC central gain changes by -0.260. These are descriptive differences, not additional significance tests. This bracket includes an illustrative or invalid endpoint and is not established. The interval brackets sampled tolerance labels only; it does not locate a unique continuous temperature threshold or identify a physical failure mode. Every new temperature configuration satisfies the preselected signal and late-range checks. These checks bound the measured late variation, but do not prove equilibrium of the distribution functions.

Prospective temporal test. The two-sided tests use the within-seed difference between the two late times after averaging the three windows. The independent units are the three base seeds, giving two degrees of freedom per one-sample t test. The normal sampling assumption is difficult to assess with three realizations, so the result is conditional on that model. For comparison, a two-sided exact sign test with three observations has minimum attainable $p = 2/2^3 = 0.25$; the finer resolution of the reported t tests relies on their sampling assumption. This resolution bound is not an additional executed test. The two originally planned tests and the additional fresh refinement test join the three untestable original physical/validity entries in one six-member Holm family at level 0.05. The latter entries use $p = 1$ only for accounting; they are not measured probabilities. The additional test was fixed before observing its three independent fresh realizations; its inclusion enlarges the original family conservatively. The corrected tests detect late evolution at 0.455 keV. Thus finite-time nonstationarity is directly supported at those new configurations. Its causal contribution to the residual is not isolated by this test: neither the first failing physical approximation nor the fraction of the gain error caused by temporal evolution is identified.

Spatial integration versus late-time variation. At 1.2 keV, the mean transverse-window contrast after averaging the two late times is 0.1665, compared with a mean absolute within-seed temporal drift of 0.0143. Appendix C gives the same decomposition for every configuration. These are descriptive contrasts, not independent variance components. They distinguish a stable average trace from robustness to the transverse integration aperture; neither substitutes for equilibrium or numerical convergence.

Measured state and diagnostic sensitivity. In the completed zero-field coarse reference at 3.600 ps, the core moments are $T_e = 553.2$ eV, $T_i = 216.9$ eV, and $n_e/n_{e0} = 0.9531$. Replacing the input Maxwellian moments by these independently measured moments changes the stationary prediction from 0.0117 to 0.0558, while the primary PIC gain is 0.6609. This diagnostic was specified after inspecting the archived spectral mismatch but before reading the new temperatures. It uses no gain fit and does not replace the primary baseline. The deposited-temperature estimator has a separate initialized offset, audited in Appendix C.1; the substitution does not correct that offset. Moments alone do not establish a Maxwellian distribution, and a homogeneous moment substitution cannot reproduce spatial evolution or a causal finite-time response. A further crossed sensitivity calculation combines those measured moments with cold boundary-phase matching or the independently registered optical beat vector. The corresponding stationary gains are 0.0956 and 0.0769, respectively (Appendix E). All input combinations are retained; neither substitution replaces the primary result or resolves the zero-field discrepancy.

Velocity distributions at the adaptive point. Across its 3 fresh realizations, the ion global velocity-covariance anisotropy spans 1.0045 to 1.0048 at initialization and 1.436 to 1.442 at 3.6 ps. The late ion marginal CDF distance from its moment-matched Gaussian spans 0.0092 to 0.0097. Electron global velocity-moment temperatures change from 452.6 to 452.9 to 390.3 to 390.7 eV. These are particle-covariance diagnostics, distinct from the deposited temperatures above; Appendix F retains every seed, species, and snapshot. Gain compatibility therefore coexists with changed velocity moments and ion marginal shape. The global covariance includes spatial bulk variation and coherent quiver motion. These descriptive changes do not isolate the cause of a temperature boundary, and snapshots without trajectories do not establish trapping. No particle-level significance test is used.

Computational cost. The 12 completed full-duration EPOCH runs consume 7.29 summed wall-clock hours with 16 MPI ranks; their median duration is 36.09 minutes. This excludes setup builds and aborted I/O tests, which are recorded separately, and does not estimate the unknown cost of the archived simulations. The archive-wide kinetic evaluation takes 4.52 s on one process. These are observed wall times in the shared execution environment, not isolated performance benchmarks. Five additional PIC runs would cost approximately 3.01 hours at the observed median rate; this is a projection, not an executed experiment. In this design, two runs could complete three-seed support at 0.8 keV, and three could sample one further intermediate temperature. That would refine a tolerance bracket while leaving the equilibrium and numerical-convergence limitations to be addressed by dedicated controls.

Compactness including the new calculations. A descriptive repetition of the fixed-ridge comparison includes all 76 archived and new configuration-level observations. Matching physical inputs across different box lengths are held out together, giving 73 folds; seeds are pooled before this analysis. The additional temperature is represented as a categorical input, and the penalty remains one. The best pooled predictor uses field azimuth, with mean squared error 0.1091 versus 0.1092 for the intercept alone (ratio 0.999). The original archive-only comparison remains separately reported. The sparse, adaptively augmented temperature line and the different interaction lengths limit any population-level interpretation of this predictive audit; it does not establish that an omitted physical variable is inert.

7 Implications for using reduced kinetic models

The map answers a narrower but necessary practical question: where is the stationary reconstruction numerically compatible with the measured finite-plasma gain, given the available sensitivity information? It does not establish a universal field at which Vlasov physics becomes unreliable. In particular, suppression toward zero gain can move a point inside an absolute tolerance without providing strong evidence for the drive coefficient. Quantitative use in an implosion model requires a validated unmagnetized reference at the local plasma state and an interaction-time and geometry treatment consistent with that reference.

The distinction has concrete consequences for model selection. The trends of polarization mixing and magnetic suppression can inform qualitative reasoning, consistent with magnetized LPI theory (10; 6; 8). The present carbon normalization and line shape do not justify an absolute transfer coefficient at the initial equilibrium parameters. A finite-time kinetic or PIC calculation is appropriate

when the local distributions, losses, or nonlinear evolution must be resolved; using more expensive PIC does not by itself remove the need for spatial and particle convergence. Similarly, an empirical CBET clamp calibrated inside an integrated magnetized-implosion calculation (17) cannot independently validate the homogeneous coupling model.

Several limitations restrict extrapolation. The baseline uses a spherical phase average with cold high-frequency polarizations, while the measurements come from an elongated open box with focused injection. Neither thermal high-frequency corrections nor the exact boundary-value problem is fully reproduced. The nominal propagation directions also omit refraction and numerical phase shifts at injection; the raw-field diagnostics in Appendix E quantify the resulting optical-wavevector difference. The archive includes only a small number of temperatures, different species and numerical settings, and a single source realization per parameter setting. Its documented ranges are sensitivity measures. The tolerance and low-gain floor are operational design choices, not universal accuracy requirements for every ICF application. No intensity-controlled experiment identifies nonlinear ion drive, and the nearest sound-frequency root does not enumerate all magnetized branches. These restrictions are reasons to withhold a physical attribution, not to conceal the numerical residuals.

The reproducible finding is that a gain-disagreement map and a kinetic-validity boundary are different objects. Here a major mismatch is already present before the magnetic perturbation is introduced, and neither the archive-wide compactness analysis nor the prespecified physical mechanisms supplies a unique missing-physics explanation. The fixed-geometry carbon line brackets sampled changes between compatible and discrepant labels over 0.4 to 0.455 keV, with three independent seeds at every definitive endpoint. The targeted 0.455 keV refinement was selected adaptively before its independent realizations. These are tolerance brackets with unattributed physical mechanisms, not confidence bounds on unique critical temperatures. A useful next comparison would hold species, intensity, numerical convergence, and local equilibrium fixed while resolving the causal temporal response; merely adding isolated field points would leave the central identification problem intact.

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A Reconstruction details and numerical checks

The public author manuscript of Moody and Shi (8) was obtained from OSTI record 3818101. Its repository cover is dated September 2026, while its internal manuscript date is November 2025. The publisher metadata establishes the 2026 journal publication, but we do not assume that every equation in this accessible version is identical to the final typeset version. The implementation is derived from the driven-current equation and solves the coupled modal system directly, rather than relying on a transcribed closed-form inverse.

Susceptibility tensor. Choose an orthonormal basis with $\mathbf{q} = (q_\perp, 0, q_\parallel)$ and $q_\parallel > 0$; when reversing the field-axis basis to obtain this convention, retain the corresponding sign of the physical gyrofrequency. Write $\Gamma_n = e^{-b} I_n(b)$, $\dot{\Gamma}_n = e^{-b} (I'_n - I_n)$, $\zeta_n = (\omega - n\Omega)/(q_\parallel v)$, and $Z_n = Z(\zeta_n)$. For a species, define

$$\begin{aligned} A_{xx} &= \sum_n (n^2/b) \Gamma_n Z_n, & A_{xy} &= i \sum_n n \dot{\Gamma}_n Z_n, \\ A_{xz} &= -\text{sgn}(\Omega) \sum_n \frac{n \Gamma_n Z'_n}{\sqrt{2b}}, & A_{yy} &= \sum_n \left(\frac{n^2}{b} \Gamma_n - 2b \dot{\Gamma}_n \right) Z_n, \\ A_{yz} &= i \text{sgn}(\Omega) \sqrt{b/2} \sum_n \dot{\Gamma}_n Z'_n, & A_{zz} &= -\sum_n \Gamma_n \zeta_n Z'_n. \end{aligned} \quad (6)$$

Here $Z'(z) = -2[1 + zZ(z)]$, evaluated with eight differentiated asymptotic terms when $|z| > 30$ and $|\text{Im } z| < 0.2|\text{Re } z|$ to avoid subtractive cancellation, $A_{yx} = -A_{xy}$, $A_{zx} = A_{xz}$, and $A_{zy} = -A_{yz}$. Then $\chi = \omega_{pa}^2 A/(\omega q_\parallel v)$, rotated back into the laboratory basis. The exactly parallel and unmagnetized limits are evaluated analytically to avoid the apparent singularities in these component expressions. Tensor projection, the unmagnetized full/electrostatic identity, harmonic truncation, and the energy integral all have separate numerical checks; none uses PIC gain as an expected answer. A separate cold-fluid limit check at a test temperature of 10^{-6} eV agrees with the cold quiver tensor to 1.88×10^{-6} relative error. The initial direct derivative formula lost precision in that extreme limit; the stable evaluation changes the largest archived gain by only 1.12×10^{-13} and changes no classification. The previous solver, failed limit check, and original sweep are retained. These extreme-temperature arrays are numerical fixtures, not simulated plasma observations.

Electromagnetic modes and coupling. Let $M(\omega)$ be the cold-electron quiver matrix, $M_{\perp\perp} = (I_\perp - i r [\hat{\mathbf{b}} \times]) / (1 - r^2)$ with $r = \Omega_e/\omega$, and $M_{\parallel\parallel} = 1$. The high-frequency dielectric is $\epsilon = 1 - \sum_a (\omega_{pa}^2/\omega^2) M_a$. Eliminating the field along the propagation direction leaves a Hermitian two-dimensional eigenproblem for n_j^2 . The maximum full Maxwell-equation residual is 5.25×10^{-16} . The injected linear polarization determines modal amplitudes a_{pj} and a_{sj} by projection onto transverse eigenvectors $\mathbf{e}_{pj}$ and $\mathbf{e}_{sj}$.

For reproducibility, define $S_{uv} = S(|\mathbf{k}_u - \mathbf{k}_v|L/2)$ and use the analogous notation for a sum of wavevectors. Let t label the projection equation, m a seed mode, and n a pump mode; the remaining summed indices label the beam indicated by their subscript. Then the two-by-two system is

$$\begin{aligned} \sum_j C_{tj} \delta n_j &= b_t, & C_{tj} &= n_{sj} |a_{sj}|^2 S_{sj,st}^2, \\ b_t &= -\frac{I_p}{16n_e m_e c \omega_p \omega_s} \sum_{mn} a_{sm} a_{pn}^* \mathcal{R}_{mn}^* P_{mn} H_{tmn}, \\ H_{tmn} &= \sum_{ij} a_{si}^* a_{pj} (e_{si}^\dagger M_p e_{pj}) S_{si,st} S_{sm-pn-st+pj}, \\ P_{mn} &= \mathbf{e}_{sm}^\dagger \chi_{ec} \mathbf{e}_{pn}^* + \mathbf{e}_{pn}^\dagger \chi_{ec} \mathbf{e}_{sm}. \end{aligned} \quad (7)$$

The intensity I_p in this equation is in W m^{-2} . The response $\mathcal{R}_{mn}$ uses $\mathbf{q} = \mathbf{k}_{pn} - \mathbf{k}_{sm}$ and positive beat frequency. The cold susceptibility χ_{ec} is evaluated at the mean laser frequency. The expression is the spherical interaction average of the driven-current equation; the two coupled index increments are not independently fitted gains. Direct linear solution is used unless the condition number exceeds

10^{12} , when a fixed-tolerance least-squares limit is used. The largest condition number in the archive evaluation is 903.8.

At zero field, the direct limiting expression is

$$\delta n_s = \frac{I_p q^2}{8n_e m_e c \omega_p \omega_s n_s} \frac{\omega_{pe}^2}{\bar{\omega}^2} \left[\frac{\chi_{eL}(1 + \chi_{iL})}{1 + \chi_{eL} + \chi_{iL}} \right]^* |\mathbf{e}_p^* \cdot \mathbf{e}_s|^2. \quad (8)$$

The tensor implementation recovers this limit, and orthogonal unmagnetized transfer is exactly zero. These identities verify implementation consistency; they do not validate the finite-boundary or stationary assumptions against PIC.

Finite-box sensitivity. Following `Gain_Ana2Num.m`, the real pump-on seed wavevector determines adjacent modes separated by $2\pi/L$. Amplitude fractions $f_{\pm}$ interpolate that wavevector, and each adjacent linear electromagnetic mode is mapped back to its frequency. Equation (4) is evaluated at both frequencies. The central estimate is

$$g_{\text{box}} = \log \frac{f_+^2 F(G_+) + f_-^2 F(G_-)}{f_+^2 + f_-^2}. \quad (9)$$

The common interference contribution to numerator and denominator is bounded by $2/(L|k_R^+ - k_R^-|)$, also respecting $|\langle \cos \rangle| \leq 1$. The logged bound never reaches unity in this evaluation. This borrowed prescription is explicitly a sensitivity analysis, not a newly derived open-boundary response. The electrostatic-only calculation is another labeled ablation; neither replaces the primary tensor result when it agrees better.

Measurement port. The two-dimensional filter has paired plateaus centered on $(\pm k_0, 0)$, full longitudinal width $0.05k_0$, full transverse width $2\pi/L_y$, and Gaussian tails with those same standard deviations. After integrating across each transverse fraction, the peak is taken on the positive-frequency half of the real signal’s Fourier spectrum; this removes a numerical tie between conjugate peaks. A second one-dimensional filter is applied. The real E_z array is shifted by $\psi/\bar{k}$ using linear interpolation and the unverifiable boundary samples are discarded, matching the supplied shift routine. The resulting orthogonal combinations yield modal energies. A synthetic amplitude-scaling check, explicitly separate from PIC data, recovers a known log-energy change to 5.16×10^{-15} absolute error. The archived hydrogen Fourier-power estimator is not silently reinterpreted as this real-space measurement.

B Complete configuration-level results

All entries are at $\Delta\lambda = 0.9 \text{ \AA}$. In the parameter tuple $(B, \phi_B, S, \text{pol.})$, B is in MG and ϕ_B in degrees. C denotes carbon with $(n_e, T_e, I) = (3.5 \times 10^{20} \text{ cm}^{-3}, 0.8 \text{ keV}, 5 \times 10^{14} \text{ W cm}^{-2})$; H_4 and H_{12} denote hydrogen with $(n_e, I) = (3 \times 10^{20} \text{ cm}^{-3}, 10^{15} \text{ W cm}^{-2})$ and $T_e = 0.4, 1.2 \text{ keV}$, respectively. Thus each tuple specifies all six comparison parameters. Geometry and numerical differences are given in Sec. 3. A dagger marks illustrative support. The plus/minus quantity is the full documented sensitivity range, not a standard error. C, D, and U mean compatible, discrepant, and unresolved against the reconstructed baseline. “None” means mechanism-unattributed, not absence of a mechanism. New-run rows R denote numerical references and N denote the temperature configurations. In these rows C_j^r has the same carbon density and intensity, $T_e = j/10 \text{ keV}$, $T_i = T_e/4$, and $L = 250\lambda_0$ rather than the archived $500\lambda_0$. Repeated reference tuples, if present, are the declared coarse and fine attempts, in execution order. A reference that fails numerical validation remains reported but does not establish a physical boundary. A double dagger identifies the adaptive temperature refinement, evaluated with independent fresh seeds. The temperature rows pool every retained seed and window/time sample; their validity checks are given in Sec. 6. I(C), I(D), or I(U) marks a failed measurement prerequisite with the underlying numerical label retained in parentheses.

Table 4: Every archived configuration and all completed new configurations, with no residual-dependent exclusions.

Cell	$(B, \phi_B, S, \text{pol.})$	g_V	$g_P \pm u$	D	Class	Mechanism
1	(2, -13, C, zy)	-0.000	-0.196 ± 0.200	0.391	U	None
2	(5, -13, C, zy)	0.000	-0.083 ± 0.142	0.166	U	None
3	(10, -13, C, zy)	0.003	-0.218 ± 0.224	0.443	U	None
4	(50, -13, C, zy)	0.132	-0.097 ± 0.259	0.458	U	None
5	(80, -13, C, zy)	0.122	-0.068 ± 0.183	0.381	U	None
6	(2, -13, C, zz)	0.051	1.030 ± 0.262	0.950	D	None
7	(5, -13, C, zz)	0.114	0.732 ± 0.183	0.844	D	None
8	(10, -13, C, zz)	0.189	0.125 ± 0.299	0.128	U	None
9	(50, -13, C, zz)	0.116	-0.039 ± 0.262	0.311	U	None
10	(2, 30, C, zy)	-0.000	-0.185 ± 0.227	0.369	U	None
11	(5, 30, C, zy)	-0.000	-0.024 ± 0.154	0.048	U	None
12	(10, 30, C, zy)	-0.000	0.320 ± 0.156	0.641	D	None
13	(50, 30, C, zy)	0.018	0.468 ± 0.162	0.899	D	None
14	(2, 30, C, zz)	0.024	1.039 ± 0.271	0.977	D	None
15	(5, 30, C, zz)	0.024	0.944 ± 0.284	0.974	D	None
16	(10, 30, C, zz)	0.026	0.722 ± 0.298	0.964	D	None
17	(50, 30, C, zz)	0.017	0.493 ± 0.200	0.953	D	None
18	(0, 75, C, zy)	0.000	-0.190 ± 0.771	0.380	U	None
19	(2, 75, C, zy)	0.000	-0.394 ± 0.437	0.789	U	None
20	(5, 75, C, zy)	0.000	-0.240 ± 0.228	0.480	U	None
21	(10, 75, C, zy)	0.000	-0.136 ± 0.201	0.274	U	None
22	(50, 75, C, zy)	0.009	0.501 ± 0.155	0.982	D	None
23	(0, 75, C, zz)	0.023	1.121 ± 0.266	0.979	D	None
24	(2, 75, C, zz)	0.023	1.073 ± 0.259	0.978	D	None
25	(5, 75, C, zz)	0.023	1.030 ± 0.266	0.977	D	None
26	(10, 75, C, zz)	0.023	1.026 ± 0.285	0.978	D	None
27	(50, 75, C, zz)	0.014	0.533 ± 0.364	0.974	D	None
28 [†]	(0, 15, H_4, zy)	0.000	-0.043 ± 0.000	0.085	C	None
29 [†]	(2.5, 15, H_4, zy)	-0.000	-0.024 ± 0.000	0.048	C	None
30 [†]	(5, 15, H_4, zy)	-0.001	-0.021 ± 0.000	0.040	C	None
31 [†]	(7.5, 15, H_4, zy)	-0.002	0.175 ± 0.000	0.356	D	None
32 [†]	(10, 15, H_4, zy)	-0.004	0.389 ± 0.000	0.785	D	None
33 [†]	(12.5, 15, H_4, zy)	-0.005	0.440 ± 0.000	0.889	D	None
34 [†]	(15, 15, H_4, zy)	-0.005	0.597 ± 0.000	1.008	D	None
35 [†]	(17.5, 15, H_4, zy)	-0.003	0.614 ± 0.000	1.005	D	None
36 [†]	(20, 15, H_4, zy)	0.001	0.648 ± 0.000	0.999	D	None
37 [†]	(22.5, 15, H_4, zy)	0.008	0.627 ± 0.000	0.987	D	None
38 [†]	(25, 15, H_4, zy)	0.018	0.734 ± 0.000	0.975	D	None
39 [†]	(27.5, 15, H_4, zy)	0.031	0.670 ± 0.000	0.954	D	None
40 [†]	(30, 15, H_4, zy)	0.045	0.724 ± 0.000	0.938	D	None
41 [†]	(0, 15, H_4, zz)	0.555	2.073 ± 0.000	0.732	D	None
42 [†]	(2.5, 15, H_4, zz)	0.557	1.950 ± 0.000	0.714	D	None
43 [†]	(5, 15, H_4, zz)	0.570	1.787 ± 0.000	0.681	D	None
44 [†]	(7.5, 15, H_4, zz)	0.577	1.621 ± 0.000	0.644	D	None
45 [†]	(10, 15, H_4, zz)	0.580	1.339 ± 0.000	0.566	D	None
46 [†]	(12.5, 15, H_4, zz)	0.582	1.356 ± 0.000	0.571	D	None
47 [†]	(15, 15, H_4, zz)	0.581	1.014 ± 0.000	0.427	D	None
48 [†]	(17.5, 15, H_4, zz)	0.578	1.017 ± 0.000	0.431	D	None
49 [†]	(20, 15, H_4, zz)	0.572	0.991 ± 0.000	0.423	D	None
50 [†]	(22.5, 15, H_4, zz)	0.562	0.799 ± 0.000	0.296	C	None
51 [†]	(25, 15, H_4, zz)	0.549	0.721 ± 0.000	0.239	C	None
52 [†]	(27.5, 15, H_4, zz)	0.533	0.651 ± 0.000	0.182	C	None
53 [†]	(30, 15, H_4, zz)	0.514	0.734 ± 0.000	0.300	C	None
54	(0, 15, H_{12}, zy)	0.000	-0.184 ± 0.038	0.368	U	None
55	(2.5, 15, H_{12}, zy)	-0.000	-0.155 ± 0.039	0.310	U	None
56	(5, 15, H_{12}, zy)	-0.000	-0.081 ± 0.044	0.162	C	None
57	(7.5, 15, H_{12}, zy)	-0.000	-0.091 ± 0.121	0.181	U	None

Table 4 continued

Cell	$(B, \phi_B, S, \text{pol.})$	g_V	$g_P \pm u$	D	Class	Mechanism
58	(10, 15, H_{12}, zy)	-0.001	0.045 ± 0.045	0.092	C	None
59	(12.5, 15, H_{12}, zy)	-0.001	0.050 ± 0.079	0.102	C	None
60	(17.5, 15, H_{12}, zy)	-0.001	0.188 ± 0.099	0.377	U	None
61	(20, 15, H_{12}, zy)	-0.000	0.232 ± 0.053	0.463	D	None
62	(0, 15, H_{12}, zz)	0.068	0.629 ± 0.066	0.892	D	None
63	(2.5, 15, H_{12}, zz)	0.068	0.601 ± 0.133	0.887	D	None
64	(5, 15, H_{12}, zz)	0.068	0.609 ± 0.114	0.887	D	None
65	(7.5, 15, H_{12}, zz)	0.070	0.534 ± 0.145	0.869	D	None
66	(10, 15, H_{12}, zz)	0.071	0.513 ± 0.054	0.861	D	None
67	(12.5, 15, H_{12}, zz)	0.072	0.308 ± 0.046	0.473	D	None
68	(15, 15, H_{12}, zz)	0.072	0.451 ± 0.044	0.759	D	None
69	(17.5, 15, H_{12}, zz)	0.072	0.188 ± 0.153	0.233	U	None
70	(20, 15, H_{12}, zz)	0.071	0.242 ± 0.032	0.343	U	None
R1 [†]	(0, 75, C_8^r, zz)	0.012	0.661 ± 0.218	0.982	D	None
R2 [†]	(50, 75, C_8^r, zz)	0.011	0.546 ± 0.201	0.981	D	None
N1	(10, 30, C_4^r, zz)	1.534	0.874 ± 0.149	0.755	D	None
N2 [‡]	(10, 30, $C_{4,55}^r, zz$)	0.794	0.819 ± 0.172	0.030	C	None
N3 [†]	(10, 30, C_8^r, zz)	0.013	0.559 ± 0.195	0.977	D	None
N4	(10, 30, C_{12}^r, zz)	0.008	0.283 ± 0.256	0.549	U	None
Compactness: archive		Intercept is best; LOO MSE = 0.1068.				
Carbon z - z subset		B, ϕ_B : MSE = 0.0629 (27.2% reduction).				

C Individual realizations and numerical scales

Every completed full-duration realization appears below. The two reference fields and the temperature control are single-seed diagnostics. The original endpoint temperatures use the same three declared base seeds, while the adaptive refinement uses a separate set. Every realization is retained without selection. The late drift is $g(3.6 \text{ ps}) - g(3.0 \text{ ps})$ after averaging the three windows. The “range” is over the six time/window measurements within a realization. These observations are also available separately in the raw SDF and analysis logs.

Configuration	Seed	g_P	Range	Late drift	SNR	Minutes
C_T1200 (12)	7842432	0.3088	0.1941	-0.0032	113.1	36.22
C_T1200 (12)	7842533	0.2587	0.1951	-0.0258	114.1	36.42
C_T1200 (12)	7842634	0.2804	0.1939	0.0138	111.9	35.13
C_T400 (12)	7842432	0.8548	0.1441	0.1059	172.9	34.39
C_T400 (12)	7842533	0.8823	0.0684	0.0279	172.4	36.10
C_T400 (12)	7842634	0.8844	0.0882	0.0500	178.9	33.10
C_T455 (12)	7952432	0.8480	0.1099	0.0617	182.5	33.73
C_T455 (12)	7952533	0.8120	0.1213	0.0731	174.6	36.08
C_T455 (12)	7952634	0.7961	0.1276	0.0798	171.4	33.27
C_T800_control (12)	7842432	0.5591	0.1947	0.0926	151.0	36.21
V_B0 (12)	7842432	0.6609	0.2175	0.1193	154.7	50.26
V_B50 (12)	7842432	0.5455	0.2009	0.0923	146.5	36.25

The configuration suffix gives cells per wavelength and total particles per cell. The 16 MPI ranks use the base seed plus rank. This assigns distinct declared initializations; it does not imply exact bitwise equivalence across a different MPI decomposition. All comparisons here keep the decomposition fixed.

Resolution limits. Across the completed runs, the initial mesh-to-electron-Debye-length ratio lies between 2.12 and 3.68, $\omega_{pe} \Delta t$ equals 0.0692 throughout, and $|\Omega_e| \Delta t$ lies between 0.0000 and 0.0576. The Debye scale is therefore not spatially resolved by multiple cells. Reference amplitude agreement under the declared tolerance is a limited calibration, not a proof of spatial/particle convergence or of negligible finite-grid heating. The new temperature comparisons inherit this limitation even when their late-time spectral signal is strong. The timestep diagnostic uses the two-dimensional Courant

expression and the input multiplier, and is distinct from a measured dispersion-error bound. The largest relative difference between the sum of the two extracted modal energies and the unshifted transverse-field energy is 0.0150. The spatial phase interpolation changes the sampled norm slightly, so these are not forced to match. The primary total-energy gain uses the unshifted transverse-field norm, consistently with the supplied archive’s branch-zero convention.

Measured-state sensitivity is secondary. The following table gives each realization’s late core electron temperature and density and the corresponding homogeneous Maxwellian prediction. The diagnostic changes only moments measured independently from the raw particles’ deposited fields. It cannot identify a distribution function from its first moments or distinguish physical cooling from numerical effects. The deposited estimator also has an initialized offset, audited in Appendix C.1; no empirical bias correction is applied. All primary residuals still use the initial input state.

Configuration	Seed	T_e (eV)	T_i (eV)	n_e/n_{e0}	Moment-state g_V
C_T1200	7842432	698.3	319.0	0.9198	0.0372
C_T1200	7842533	698.4	319.2	0.9200	0.0372
C_T1200	7842634	698.5	319.4	0.9200	0.0372
C_T400	7842432	339.9	116.3	0.9795	0.1746
C_T400	7842533	339.4	116.4	0.9796	0.1730
C_T400	7842634	339.9	116.5	0.9797	0.1755
C_T455	7952432	368.9	130.7	0.9753	0.8228
C_T455	7952533	369.1	130.8	0.9753	0.8335
C_T455	7952634	369.3	131.1	0.9752	0.8549
C_T800_control	7842432	532.9	219.7	0.9484	0.0752
V_B0	7842432	553.2	216.9	0.9531	0.0558
V_B50	7842432	512.4	254.0	0.9510	0.0882

The spatially low-pass longitudinal electric-field RMS and a pump-side spectral-energy ratio are also logged. The field diagnostic retains $|k_x|, |k_y| < 0.2k_0$ and averages over the full box; it does not isolate a sheath by temporal filtering. The pump ratio compares two spatial strips and is explicitly a proxy, not a directional Poynting-flux balance or a causal pump-depletion measurement. No pump-depletion attribution is inferred from it.

C.1 Initialized temperature estimators

Inspection of EPOCH’s `calc_temperature` motivated a further descriptive audit after the original endpoint runs, before the adaptive realizations. Its deposited temperature subtracts a local mean momentum and divides the deposited momentum variance by the particle count and three degrees of freedom. It uses no finite-sample variance correction. This differs from the global core velocity covariance in Appendix F, which includes spatial variation of bulk motion and converts momentum relativistically. Consequently a difference from the nominal input temperature at initialization must not be called subsequent cooling.

We read the initialized SDF for every completed full-duration run, using the same density weighting and core mask as the original state diagnostic. Every row below has measured initialization, transit, and final snapshots. The table gives the initialized deposited-to-input ratio and the final-to-initial ratio of that same deposited estimator. Neither ratio corrects the input state or the primary model. Finite-particle estimation, local versus global bulk subtraction, and real evolution are distinct issues; these checks do not replace a resolution study.

Configuration	Seed	$T_{e0}^d/T_{e,in}$	$T_{i0}^d/T_{i,in}$	T_{e2}^d/T_{e0}^d	T_{i2}^d/T_{i0}^d
C_T1200	7842432	0.9583	0.9373	0.6072	1.1344
C_T1200	7842533	0.9582	0.9377	0.6074	1.1346
C_T1200	7842634	0.9582	0.9384	0.6074	1.1345
C_T400	7842432	0.9583	0.9373	0.8866	1.2406
C_T400	7842533	0.9582	0.9377	0.8857	1.2413
C_T400	7842634	0.9582	0.9384	0.8868	1.2409
C_T455	7952432	0.9571	0.9367	0.8472	1.2266
C_T455	7952533	0.9578	0.9380	0.8468	1.2260
C_T455	7952634	0.9574	0.9387	0.8478	1.2274
C_T800_control	7842432	0.9583	0.9373	0.6951	1.1720
V_B0	7842432	0.9583	0.9373	0.7216	1.1574
V_B50	7842432	0.9583	0.9373	0.6683	1.3549

Here superscript d denotes the deposited diagnostic, subscript 0 initialization, and subscript 2 the final 3.6 ps snapshot. Independently measured particle-covariance moments from the full adaptive realizations verify the loading separately from the deposited estimator. All absolute temperatures, transit values, and source hashes remain in the diagnostic log. The original measured-state sensitivity predictions use the deposited moments without bias correction, so they remain exploratory input substitutions rather than exact reconstructions of the evolving equilibrium. At initialization in the first full 0.455 keV realization (seed 7952432), the deposited electron and ion temperatures are 435.47 and 106.55 eV, whereas the global particle-velocity moments are 452.62 and 113.58 eV, respectively, for inputs 455 and 113.75 eV.

Separating time and transverse-window variation. For each seed, write the two-time, three-window measurement as

$$g_{tf} = \bar{g} + (\bar{g}_t - \bar{g}) + (\bar{g}_f - \bar{g}) + r_{tf}. \quad (10)$$

The time contrast is $\delta_t = \bar{g}_2 - \bar{g}_1$, the transverse contrast is $W = \max_f \bar{g}_f - \min_f \bar{g}_f$, and the interaction contrast is $J = \max_{tf} r_{tf} - \min_{tf} r_{tf}$. The following quantities average the indicated nonnegative contrasts over all seeds at a configuration. This algebraic diagnostic was added after the first 1.2 keV realization and is reported for every configuration. It does not replace the full pooled sensitivity range or treat windows as independent samples.

Configuration	Seeds	Mean $ \delta_t $	Mean W	Mean J
C_T1200	3	0.0143	0.1665	0.0287
C_T400	3	0.0613	0.0390	0.0097
C_T455	3	0.0715	0.0481	0.0101
C_T800_control	1	0.0926	0.1027	0.0200
V_B0	1	0.1193	0.0987	0.0173
V_B50	1	0.0923	0.1090	0.0184

The contrasts are not additive variance fractions. In particular, a small window-averaged temporal change can coexist with a large transverse-window sensitivity. A flat mean-gain trace therefore does not establish that the measured gain is insensitive to spatial integration, or that the underlying particle distributions are stationary.

D Mechanism evidence at the sampled transitions

The endpoint audit below applies the same prespecified candidate mechanisms to each observed tolerance bracket. For the drive approximation, both sides of its two-frequency ordering are shown. For weakly damped mediation, $\gamma = |\text{Im } \omega_{\text{root}}|$ comes from the fixed sound-frequency root selector at the initial state, and γt_2 uses $t_2 = 3.6$ ps. Neither this selector nor a change in its output enumerates all magnetized branches. The ion gyrophase product uses the same full simulation time, an upper bound on the elapsed pump-driven interaction because pump injection is delayed. These time products describe relaxation and orbital timescales rather than supply sharp validity cutoffs. The nominal propagation angles remain in the accompanying log. The tabulated quantities are model diagnostics; only the tolerance labels use measured gain.

Endpoint	Label	$ \Omega_e /\omega_p$	$\Delta\omega/ \Omega_e $	γt_2	$ \Omega_i t_2$
H_{12} , zy, 12.5 MG	C	0.0410	0.0063	1.78	0.431
H_{12} , zy, 20 MG	D	0.0655	0.0039	1.85	0.690
H_4 , zy, 5 MG [†]	C	0.0164	0.0156	0.978	0.172
H_4 , zy, 7.5 MG [†]	D	0.0246	0.0104	1.02	0.259
H_4 , zz, 20 MG [†]	D	0.0655	0.0039	1.12	0.690
H_4 , zz, 22.5 MG [†]	C	0.0737	0.0035	1.14	0.776
C , 0.4 keV	D	0.0328	0.0078	0.0592	0.172
C , 0.455 keV	C	0.0328	0.0078	0.0623	0.172
C , 0.8 keV [†]	D	0.0328	0.0078	0.0769	0.172

Here H_4 , H_{12} denote the archived hydrogen temperatures and C the new fixed-geometry carbon line. A dagger marks illustrative support. Every endpoint in this table has a successfully recovered root retained in the solver log. Frequency separation is not converted into an empirical error threshold. The intensity and field direction are fixed within each bracket, preventing an independent test of nonlinear ion drive or an angle-controlled causal comparison.

At the common detuning, the imposed beat period is 4.567 ps, longer than the 3.6 ps simulation duration. The delayed pump shortens the driven interval further. This temporal scale limits identification of a stationary narrow-band plasma mode from the spatial snapshots, without by itself assigning the gain residual to a specific mechanism.

Transition 1: H_{12} , zy, 12.5 MG to H_{12} , zy, 20 MG. The endpoint labels are compatible and discrepant; both endpoints have sensitivity support. The frequency-ordering and selected-root diagnostics quantify the two planned physical candidates, but the archive supplies no raw temporal, particle, or pump-flux control with which to isolate them. The nonlinear-ion-drive candidate lacks an intensity contrast. Linear harmonic resonances are already retained in the model. The first physical failure is therefore mechanism-unattributed for this bracket; a temporal rejection, when present, instead identifies a comparison-validity concern.

Transition 2: H_4 , zy, 5 MG to H_4 , zy, 7.5 MG. The endpoint labels are compatible and discrepant; at least one endpoint is illustrative or fails a measurement prerequisite. The frequency-ordering and selected-root diagnostics quantify the two planned physical candidates, but the archive supplies no raw temporal, particle, or pump-flux control with which to isolate them. The nonlinear-ion-drive candidate lacks an intensity contrast. Linear harmonic resonances are already retained in the model. The first physical failure is therefore mechanism-unattributed for this bracket; a temporal rejection, when present, instead identifies a comparison-validity concern.

Transition 3: H_4 , zz, 20 MG to H_4 , zz, 22.5 MG. The endpoint labels are discrepant and compatible; at least one endpoint is illustrative or fails a measurement prerequisite. The frequency-ordering and selected-root diagnostics quantify the two planned physical candidates, but the archive supplies no raw temporal, particle, or pump-flux control with which to isolate them. The nonlinear-ion-drive candidate lacks an intensity contrast. Linear harmonic resonances are already retained in the model. The first physical failure is therefore mechanism-unattributed for this bracket; a temporal rejection, when present, instead identifies a comparison-validity concern.

Transition 4: C , 0.4 keV to C , 0.455 keV. The endpoint labels are discrepant and compatible; both endpoints have sensitivity support. The prospective temporal controls give 0.4 keV has Holm $p = 0.5929$, and 0.455 keV has Holm $p = 0.03234$. State moments and spatial-field controls are retained at both endpoints; distribution snapshots at the adaptive point add information about shape but no causal control across temperature. The nonlinear-ion-drive candidate lacks an intensity contrast. Linear harmonic resonances are already retained in the model. The first physical failure is therefore mechanism-unattributed for this bracket; a temporal rejection, when present, instead identifies a comparison-validity concern.

Transition 5: C , 0.455 keV to C , 0.8 keV. The endpoint labels are compatible and discrepant; at least one endpoint is illustrative or fails a measurement prerequisite. The prospective temporal controls give 0.455 keV has Holm $p = 0.03234$. State moments and spatial-field controls are retained at both endpoints; distribution snapshots at the adaptive point add information about shape but no

causal control across temperature. The nonlinear-ion-drive candidate lacks an intensity contrast. Linear harmonic resonances are already retained in the model. The first physical failure is therefore mechanism-unattributed for this bracket; a temporal rejection, when present, instead identifies a comparison-validity concern.

E Spatial-field checks on mechanism attribution

The measured-state and temporal diagnostics do not directly identify particle trapping. We therefore attempted a second type of diagnostic using the raw spatial electric fields. This analysis was added after inspecting the first completed reference, before observing a completed new temperature gain; it is exploratory and contributes no additional significance test.

First, project the electric field along the nominal cold-mode beat vector $\mathbf{q}$ with a separable Hann weight on the central 80% of the domain. The single-sinusoid amplitude estimate is twice the modulus of the weighted complex coefficient. Four controls displace q_y by ± 2 and ± 3 core-domain Fourier bins. The pre-pump transit coefficient is retained. At the nominal vector, spatial phase cancellation can suppress the diagnostic even if local waves exist.

The second attempt registers the optical wavevectors directly from E_y, E_z Fourier power at the final snapshot, independently of the gain residual. A fourfold zero-padded FFT initializes a bounded continuous local maximization for each optical peak. The fixed seed search is $0.85 < k_x/k_0 < 1.10$, $|k_y/k_0| < 0.12$; the pump search is $0.65 < k_x/k_0 < 1$, $-0.7 < k_y/k_0 < -0.3$. The same fitted final beat vector is then used to project both late fields and the transit field. All nominal and registered coefficients and sidebands are retained. The component-staggering difference is at most a cell and is not corrected by a fit. Multiple magnetized optical modes and focusing also limit a single-vector description.

Configuration	Seed	Pump angle	$\Delta q_x L_x / (2\pi)$	P_q / P_{sb}	E_q / E_{q0}
C_T1200	7842432	28.32°	2.56	0.15	0.33
C_T1200	7842533	28.32°	2.55	0.06	0.39
C_T1200	7842634	28.32°	2.55	0.85	0.67
C_T400	7842432	28.31°	2.76	2.86	0.75
C_T400	7842533	28.31°	2.79	2.25	3.87
C_T400	7842634	28.31°	2.79	4.26	1.13
C_T455	7952432	28.31°	2.77	1.69	1.10
C_T455	7952533	28.31°	2.77	0.10	0.77
C_T455	7952634	28.31°	2.76	0.04	0.14
C_T800_control	7842432	28.32°	2.65	0.24	1.16
V_B0	7842432	28.31°	2.67	2.76	1.48
V_B50	7842432	28.26°	2.79	3.87	1.01

The pump angle is measured relative to $+x$, and Δq_x is the registered minus nominal beat component. The power ratio compares the final registered longitudinal coefficient with the mean of the four transverse sideband powers. It is a spatial contrast, not a sampling confidence statistic. The pre-pump coefficient can contain noise and boundary fields; its value is not set to zero by construction.

For the long interaction box, even a small optical-wavevector error can produce several cycles of projection-phase mismatch. The boundary deck fixes a tangential vacuum phase, whereas the baseline assigns a prescribed direction to a homogeneous-plasma eigenmode. Refraction, numerical dispersion, focusing, and mode mixing can all contribute to the measured difference. These observations motivate a boundary-matched comparison; they do not authorize a per-cell wavevector fit to improve the primary theoretical gain.

We also log the nominal single-wave ion bounce-frequency estimate $\omega_b = (|Ze q E_q| / m_i)^{1/2}$, the associated density-perturbation estimate, and $\omega_b / |\Omega_i|$ where the denominator is nonzero. A spatial coefficient at one time does not establish a temporally coherent ion mode or trapped particle orbits. The three projection times used here, the control coefficients, and the absence of particle trajectories limit attribution to trapping. The separate refinement snapshots in Appendix F supply distribution information at one temperature, without resolving particle orbits. Neither spatial-projection attempt therefore converts a discrepant point into an established failure of the unperturbed-orbit approximation.

Joint state and wavevector sensitivity at zero field. After observing the reference discrepancy and optical-vector difference, we specified a crossed exploratory calculation at zero field. This case avoids ambiguity from separated magnetic optical modes. For each of the initial and measured core Maxwellian moments, evaluate Eq. (8) with (a) the nominal homogeneous directions, (b) cold boundary-phase matching, and (c) the optically registered beat vector. For (b), the pump tangential component is fixed by its vacuum injection, $k_{px} = (\omega_p/c) \cos \alpha$, and the inward k_{py} follows the cold unmagnetized dispersion; the seed has $k_{sy} = 0$. The total cold pump wavenumber and the positive seed wavenumber use the same state. For (c), only the measured beat vector is substituted in the susceptibility and drive; this is not a new eigenmode solution. The intensity, length, and detuning stay fixed. All six combinations are retained at both late times.

Moments	Wavevector prescription	Stationary g
Initial	Nominal direction	0.01171
Initial	Boundary phase	0.01354
Initial	Optically registered	0.01260
Measured	Nominal direction	0.05583
Measured	Boundary phase	0.09560
Measured	Optically registered	0.07694

The table uses 3.6 ps measured moments and the final optical registration. These input substitutions are independent of the gain residual. They remain homogeneous stationary sensitivities, with no propagated uncertainty on fitted optical vectors, spatial profiles, or distribution shape. An improved numerical match would not establish the causal finite-time response or validate the same substitution in a magnetized multimode plasma. The primary comparison and every compatibility label remain unchanged.

F Velocity-distribution checks in the adaptive refinement

Particle positions, three momentum components, and macroparticle weights were additionally saved for all three completed 0.455 keV realizations at initialization, the pump-transit reference, and both late times. This diagnostic was specified after the first 0.4 keV gain but before the refinement was run. Its output blocks do not alter physical or numerical inputs or the primary gain reduction. All four registered snapshots are present for each realization, and initialized particle moments independently check the loading. The diagnostic is exploratory; it contributes no significance test.

We retain particles in the central 80% of both spatial coordinates and convert momentum to relativistic velocity. With macroparticle weights w_a , the bulk velocity and covariance are

$$\bar{\mathbf{v}} = \frac{\sum_a w_a \mathbf{v}_a}{\sum_a w_a}, \quad \Sigma = \frac{\sum_a w_a (\mathbf{v}_a - \bar{\mathbf{v}})(\mathbf{v}_a - \bar{\mathbf{v}})^T}{\sum_a w_a}. \quad (11)$$

The directional moment temperatures are $T_j = m\Sigma_{jj}/e$ in eV, and the anisotropy is the ratio of the largest to smallest covariance eigenvalue. These are velocity moments; spatially varying bulk motion and coherent quiver motion are included rather than removed by an unregistered fit. Along the nominal optical beat direction, we also compare the weighted velocity CDF with a Gaussian having the same measured mean and variance. The reported supremum distance is a shape diagnostic, not a Kolmogorov–Smirnov significance test: macroparticles share self-consistent fields and are not independent replicates.

Seed	Species	T_x (eV)	T_y (eV)	T_z (eV)	Anisotropy	CDF distance
7952432	electron	381.2	382.2	407.4	1.071	0.0007
7952432	ion	155.0	163.7	113.8	1.442	0.0092
7952533	electron	381.0	382.8	407.5	1.071	0.0005
7952533	ion	154.7	163.9	114.0	1.440	0.0096
7952634	electron	381.0	383.2	408.0	1.073	0.0005
7952634	ion	155.1	163.9	114.4	1.436	0.0097

The table reports 3.6 ps; the complete initialization, transit, and late diagnostics are retained in the accompanying data. The three seeds are the independent simulation units. No uncertainty is inferred by treating millions of macroparticles as independent samples.

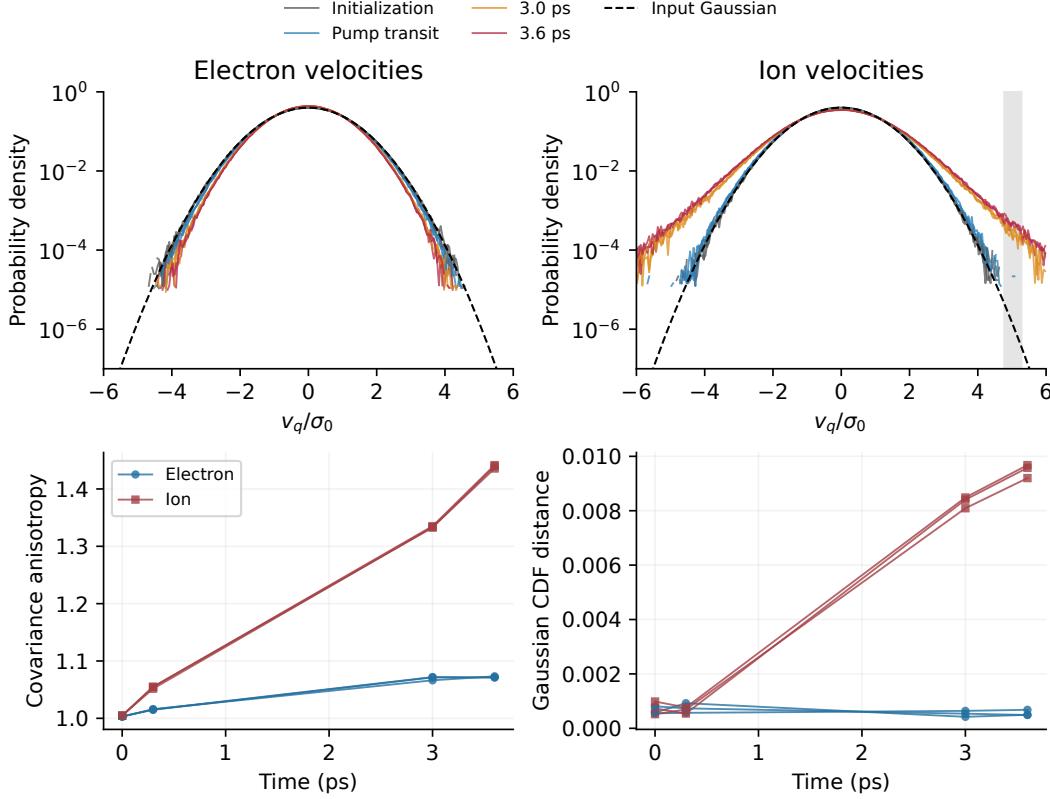


Figure 6: Every particle snapshot in the three-seed 0.455 keV refinement. Thin curves retain the individual seeds. Velocities use the initial thermal standard deviation $\sigma_0 = (T_0 e/m)^{1/2}$. The dashed curve is the input Gaussian, whereas the lower-right distances use a Gaussian with each snapshot’s measured moments. Empty histogram bins are omitted on the logarithmic axes. The shaded ion-velocity band is within $0.25\sigma_0$ of the imposed beat phase velocity. Anisotropy and non-Gaussian shape do not identify a causal physical failure of the coupling model.

The nominal beat phase speed is 5.03 initial ion thermal standard deviations. In seed order, the initialization contains 0, 2, 0 ion macroparticles in the specified phase-speed band; at 3.6 ps the counts are 275, 305, 286. The corresponding weighted fractions and total core counts are also logged. These counts describe the sampled velocity tail, not a resolved damping rate or a trapped population.

In a magnetized plasma, the cyclotron-resonance condition is $\omega - q_{\parallel} v_{\parallel} - n\Omega_i = 0$. The instantaneous phase-speed band along $\mathbf{q}$ is therefore not a census of all resonant ions, and a low count there does not imply that the retained harmonic response is absent.

Distribution snapshots strengthen the check of the initialized Maxwellian and expose subsequent changes beyond scalar deposited temperatures. They are available only at the adaptive temperature, so they cannot distinguish which feature causes a transition between temperatures. We did not attempt to infer closed trapped orbits from a folded phase portrait: particle trajectories and phase locking over multiple wave periods are not recorded. Consequently the particle and spatial-field diagnostics leave the physical mechanism of any tolerance crossing unattributed, while documenting the limits of the stationary initial-equilibrium comparison.

G Reproducibility and provenance

The accompanying repository retains the archive byte-for-byte and records its checksum before analysis. `code/sweep_plan.json` fixes the original parameter design, tolerance, hypotheses, and numerical reduction. The later temporal, input-state, middle-temperature-control, adaptive-temperature, exploratory field/geometry-diagnostic, particle-output, and initialized-temperature-estimator addi-

tions are separate dated files; the original design is not rewritten. The middle-temperature control was added after archive inspection to remove the interaction-length difference between the new carbon temperatures and the original carbon reference, before any completed new gain was read. Its single deterministic realization is illustrative. The later 0.455 keV refinement was selected after the first 0.4 keV realization, using a recorded theory-only scan; its three new seeds are disjoint from the original seeds and were fixed before its outcomes.

`logs/archive_loading.json` records all 97 gain files, including alternate estimators, their checksums, headers, and rows. `logs/evaluation.json` contains all 70 primary configuration evaluations, full available detuning curves, complex-root diagnostics, modal index increments, finite-box sensitivity estimates, and residual intervals. The CSV export and `table_row_provenance.json` map every table row to its underlying source. Source and numerical-check logs distinguish measured PIC values, reconstructed theory values, and synthetic unit-test fixtures.

All new EPOCH jobs retain their input deck, deterministic base seed, start and end timestamps, MPI command, exit status, raw SDF files, and processed snapshots in `runs/`. The random generator uses the declared base seed plus MPI rank; there is no seed selection. The only simulation-source patch exposes that base seed through an environment variable. Separately prefixed particle snapshots were added to all adaptive-refinement realizations before their execution; these files are retained alongside the original field-output schedule and audited independently. Particle loading, the pusher, field update, and laser boundaries remain the supplied EPOCH implementation (1). The initial socket-transport MPI build and the later shared-memory build use the same MPI release and EPOCH executable; the selected runtime prefix is recorded per job. A failed MPI-I/O setup was detected from absent SDF files and archived before any gain was measured.

The MATLAB measurement package was inspected and its relevant operations ported because MATLAB/Octave and its separate three-wave dependency were unavailable. The SDF reader was rebuilt from the supplied source to match the installed NumPy ABI. No external dispersion or boundary-analysis implementation was imported. The public theory manuscript, literature metadata, software build logs, inspected-code notes, and unsuccessful approaches are retained. The finite-box broadening formula, controlled-reference validation principle, Holm correction, and cross-validation all have explicit prior-work citations.

From the workspace root, the archive analysis is regenerated with

```
python proposal/code/load_archive.py
python proposal/code/check_solver.py
python proposal/code/evaluate.py
python proposal/code/summarize.py
python proposal/code/make_figures.py
python proposal/code/generate_tables.py
```

The PIC workflow scripts generate decks from the supplied example, execute the reference gate before evaluating new temperatures, and preserve failed attempts. The final artifact builder checks completed job status before regenerating the new-run section and compiling the manuscript. `codebase_notes.md` explains the reused, patched, and independently reconstructed components; `attempts_log.md` records the rationale and outcome of distinct approaches. Numerical tables and figure data are produced from the logs, not manually transcribed from the published figures.